

OpenCTI High Availability Architecture - Azure AKS

Based on Filigran HA recommendations + Azure managed services

Azure Cloud

Legend:

- Frontend
- Connectors
- Workers
- Managed
- Network

CTI Analysts / Users
HTTPS Access

Azure Load Balancer
Public IP + SSL Termination

AKS Cluster (Auto-scaling: 3-15 nodes)

ingress-nginx namespace

Nginx Ingress Controller
Replicas: 3 (HA)

openciti namespace - HA Cluster Separation

1. Frontend Platform Cluster

For users only - UI + GraphQL

OpenCTI Platform

Replicas: 3 (HA)
Image: openciti/platform
Auth: OpenID/SAML/Auth0
Resources: 4-8Gi RAM
Probes: Configured

Service: frontend

Type: ClusterIP
Port: 8080
Domain: openciti.domain.com

Users access frontend ONLY - no direct backend calls

2. Connector-Backend Cluster

For connectors - API ingestion only

OpenCTI Platform

Replicas: 3 (HA)
Image: openciti/platform
No user auth required
Resources: 4-8Gi RAM
Probes: Configured

Service: connector-api

Type: ClusterIP
Port: 8080
Domain: api.domain.com

Connectors GraphQL API RabbitMQ (async)

3. Worker-Backend Cluster

For workers - data processing only

OpenCTI Platform

Replicas: 3 (HA)
Image: openciti/platform
No user auth required
Resources: 4-8Gi RAM
Probes: Configured

Service: worker-api

Type: ClusterIP
Port: 8080
Internal only

Workers pull from RabbitMQ process ES/Redis

Workers (Separate Deployment)

OpenCTI Workers

Replicas: 9 (3x platform ratio)
Image: openciti/worker
OPENCTI_URL=worker-api
initContainers: wait for platform
Resources: 1-2Gi RAM

Connectors

MITRE, CVE, AlienVault, etc.
Replicas: 1 per connector
OPENCTI_URL=connector-api
Resources: 256Mi-512Mi RAM

Azure Managed Services (Shared by all clusters)

ElasticSearch

Azure Elastic Cloud

Redis

Azure Cache for Redis

RabbitMQ

AKS-hosted or VM

Blob Storage

Azure Blob (S3 API)

All managed services are HA by default - Multi-AZ deployment

Azure Premium SSD (Persistent Storage for AKS)

StorageClass: managed-premium | Total: ~300Gi for ES cluster only